C.1-ANNEX A — MATERIAL PRODUCTION AND WORKSHOPS

Version: V1.7
Declared alignment: Constitution V.40; Code 1 V1.12; Code 2 V1.9; Code 3 V1.13; Code 4 V1.11; Code 5 V1.7; Code 6 V1.9; Code 7 V1.10; Code 8 V1.7; Code 9 V3.4; Code 10 V1.8; Code 11 V1.6; Code 12 V1.7; Code 13 V1.9; Code 14 V1.10; Code 15 V1.13; Code 16 V1.11.

Art. 1 — Subject matter, definitions and legal nature

1. This Annex governs the material production, repair, reuse, adaptation, maintenance and management of tools and materials necessary for the Basket, health, the network and energy.
2. For the purposes of this Annex, **production** means the manufacture, repair, reuse, adaptation, maintenance, assembly, disassembly and conservation of objects, tools, components, PPE, furnishings and parts of infrastructure.
3. This Annex pursues the following objectives:
   a) to guarantee federated material and repair autonomy for the Basket and for the CBO of List A;
   b) to give priority to reuse and repair over the production of new goods;
   c) to prevent rent on essential tools and to reduce critical proprietary dependencies;
   d) to ensure the safety, traceability and reproducibility of essential production designs.
4. The following constitute inalienable infrastructural commons:
   a) the Base Workshop;
   b) the Federated Advanced Workshops;
   c) the Distributed Micro-stations;
   d) the Materials Storehouse;
   e) the inventory of essential tools.
   A produced object is a common good when it belongs to the Material Basket or is necessary for an essential function pursuant to Article 4. The other objects are goods for use managed in accordance with Code 1. The tools necessary for the production of the Basket do not constitute a commodity.
5. This Annex governs material production and maintenance. The following remain unaffected:
   a) the competences of Code 1 over the Basket, access to goods and general priorities;
   b) the competences of Code 2 over quality, STVs and the technical validation of goods;
   c) the competences of Code 6 over health risks;
   d) the competences of Code 14 over the general safety of persons, machines, tasks and environments;
   e) the competences of Codes 7 and 8 over networks and energy.

Art. 2 — Principles of material production

1. **Material sovereignty.** Every Circle 1 develops local capacity for production, maintenance and repair. The federated network is used to share competences, advanced equipment and capacities not reasonably available at local level, avoiding structural dependencies.
2. **Reuse before new production.** Before producing a new good, the Production Council verifies whether the requirement can be met through reuse, repair or adaptation. The production of a new object is permitted when reuse is not possible, is not adequate or does not guarantee the safety required.
3. **Priority of use.** The general priorities are those of Code 1, Article 32. Within production activities, capacity and materials are ordered in the following levels:
   a) level 1: goods of the Material Basket and life-saving PPE; the deferrable goods of Article 4, paragraph 1-bis, letter e), follow letter b);

b) level 2: maintenance of the critical infrastructure for health, network, energy and the Health Storehouse;

c) level 3: safety measures and tools necessary for maintenance;

d) level 4: goods and tools for the CBO of List A;

e) level 5: any other reasoned request.

In the event of scarcity, the reduction of non-essential production activities proceeds progressively from level 5 towards level 1. The reduction is resolved by the Assembly on a proposal from the Production Council and may not compromise the higher priorities defined by Code 1.

4. **Open hardware and reproducibility.** Every functionally relevant and standardised object or component produced for levels 1–4 has documentation sufficient to allow its study, repair, modification and reproduction. The documentation comprises, as necessary, the STV, drawings, the list of materials and operating procedures. For small parts that are not relevant, a reference to an accessible standard and to a replacement procedure is sufficient. Reproducibility shall in any case be guaranteed for the goods of levels 1–4.

5. **Transparency.** Every workshop keeps an up-to-date inventory of tools and materials, a maintenance register and an aggregate balance of production.

6. **Reversibility and repairability.** In design, preference is given to objects that can be disassembled, repaired and reused, with standard components where these are available and compatible with safety and quality requirements.

Art. 3 — Production Council

1. Every Circle 1 establishes a Production Council composed of three to five members with competences of at least level L2 or L3 in one or more of the following areas: manufacturing, woodworking, metalworking and mechanics, textiles, basic electronics and safety of production activities.

2. The members are drawn by lot in accordance with Article 12-bis of Code 3, remain in office for one year and may not be drawn by lot again for the same Council until twelve months after the end of their term of office. Replacement and the handling of conflicts of interest follow the same Article and Article 16 of Code 3.

3. The Production Council:

   a) looks after the Materials Storehouse and the Material Basket;

   b) prepares the technical documentation necessary for the Safety Sheets of the machines and of the production procedures;

   c) draws up the technical proposals and the production documentation necessary for the STVs, submitting them to Code 2 for the validation within its competence;

   d) manages the material balance and the production capacity;

   e) organises L1 training for the safety of workshop activities, in coordination with Codes 6 and 14;

   f) feeds the Dependency Register governed by Code 15;

   g) coordinates its activities with the Energy and Water Council of Code 8, the Network Council of Code 7 and the Health Council of Code 6, as well as with the Safety Council of Code 14.

4. The Production Council does not dispose of goods unrelated to production and does not exercise disciplinary powers. It may not access individual health data, except for the information that is strictly necessary and legitimately accessible for a specific function.

5. The Production Council is prohibited from moderating, removing or altering deliberative or political content on grounds of merit, orientation or dissent. Only the immediate technical interventions necessary to prevent a safety risk are permitted, provided that they are documented and limited to what is strictly necessary.

6. The acts of the Council are divided into:

a) **bound technical acts**, adopted in execution of an STV or a Safety Sheet already in force, documented and published without a new deliberative procedure;

b) **discretionary proposals or decisions**, subject to the consensus procedures of Code 3, after publication for at least seventy-two hours and recording of the outcome in the minutes.

Silence does not constitute consent if the proposal has not been published or if the prescribed time limit has not elapsed.

Art. 4 — Material Basket and fulfilment priorities

1. The Material Basket is a component of the Sufficiency Basket governed by Code 1. It comprises at least:

   a) the objects necessary for the conservation and cooking of food;

   b) minimum furnishings;

   c) essential linen and tableware;

   d) basic tools for domestic life;

   e) basic PPE;

   f) the components necessary for the maintenance of the functions of levels 1 and 2;

   g) essential spare parts for the microgrid and for the Network node;

   h) clothing and footwear suited to the climate and the season;

   i) personal hygiene products, including menstrual products and those for early childhood and for incontinence.

1-bis. The following rules apply to the goods of letters h) and i):

a) by way of derogation from Article 1, paragraph 4, from delivery they are personal-use goods pursuant to Article 10.3 of the Constitution;

b) Article 4 of Code 2 on the precedence of reuse and Articles 13.5 and 14.1 of the Constitution apply: washable or compostable products are preferred, and single-use products are permitted only under the conditions of Article 14.1;

c) they are procured and distributed through the Textile and Housing Storehouse provided for by Article 14 of Code 1 and are included in the annual Technical Annex of Article 18, paragraph 2, of that Code;

d) data on the withdrawal of the products of letter i) are neither published nor communicated in a form that identifies the person;

e) they are deferrable within the meaning of Article 32, paragraph 1-bis, of Code 1, except for the products that the Health Council indicates as necessary for health; in the order of paragraph 5 they follow letter b).

2. Access to the Material Basket may not be made conditional on the CBO, on the possession of specific competences or on the presence of the person in a particular place, within the limits established by Code 1. The unavailability of a good necessary for a vital function activates the urgent priority procedure provided for by the competent Codes.

3. The minimum list of the Material Basket and the related stock thresholds are defined in an STV. The thresholds are expressed in days of autonomy and safety margin; for the goods of letter h) of paragraph 1, as an allocation per person per year. The Assembly approves them on a proposal from the Production Council and obtains the opinion of the Health Council for PPE, for hygiene products and for goods intended for frail persons. For letters h) and i) of paragraph 1 the proposal lies with the Storehouse Council.

4. The list and the thresholds are reviewed at least once a year and whenever significant variations occur in the requirement, in availability or in safety conditions. The quantities and criteria are published, in compliance with safety needs.

5. Requests are fulfilled in the following order, consistently with Code 1, Article 32:

a) goods of the Material Basket and life-saving PPE; the deferrable goods of paragraph 1-bis, letter e), follow letter b);

b) maintenance of critical infrastructure;

c) safety and ordinary maintenance;

d) tools necessary for the CBO of List A;

e) further reasoned requests.

Art. 5 — STV, reuse and supply of materials

1. Every object produced or repaired for levels 1–4 conforms to an STV in force or to the applicable technical documentation. The STV defines, in relation to the good:

   a) the materials permitted and their provenance;

   b) the drawings, dimensions and tolerances;

   c) the life cycle and the maintenance requirements;

   d) the arrangements for disassembly and repair;

   e) the risks and the limits of use.

   The validation of the STV follows Code 2.

2. Preference is given, where compatible with safety and quality, to:

   a) reused or recovered raw materials and components;

   b) recycled materials;

   c) wood from sustainable local supply chains;

   d) recovered metals and fabrics;

   e) refurbished electronic components;

   f) recycled filaments for printing, where suitable.

3. For goods of level 4 or 5, documented verification of the possibility of reuse, repair or adaptation is mandatory before proceeding to the production of a new object. For levels 1–3 reuse is strongly preferred, unless it compromises health or electrical safety, as attested by the competent Council.

4. The verification is traced in the technical documentation or in the applicable STV. Self-building is promoted when it takes place with validated L2 competences, standard components and repairable and documented solutions.

5. For levels 4 and 5, the production of new objects that depend structurally on proprietary components not repairable locally is prohibited. A derogation may be granted by the Assembly of Circle 1 by qualified majority, on documented technical grounds and on the following conditions:

   a) the impossibility of an adequate alternative solution is demonstrated;

   b) the derogation has a fixed duration not exceeding twelve months;

   c) a compensatory measure favouring reuse or repairability is adopted;

   d) a binding substitution plan is approved;

   e) the plan is reviewed at least every six months.

   The derogation may not exceed twenty-four months in total, except in the case of an immediate and documented collective risk pursuant to Article 2-bis of Code 6. Every resulting critical dependency is entered in the Dependency Register in accordance with Code 15.

Art. 6 — Workshops, federated network and measurement

1. Every Circle 1 activates a Base Workshop, which may be divided into several stations, equipped at least with:

   a) hand tools for wood and light metal;

b) a station for the repair of household appliances and bicycles;

c) a station for sewing and textile work;

d) basic measuring instruments;

e) an L1/L2 maintenance bench for the Network and Energy functions;

f) a store for the Material Basket;

g) a manual stop control for the equipment, where relevant.

The fitting-out and use comply with the Safety Sheets and the requirements of Code 14.

2. The Confederation ensures the effective access of every Circle 1 to a Federated Advanced Workshop. Planning is defined for each Circle 2 by its General Assembly, on a proposal from the shared Production Council provided for by Article 12-bis, paragraph 6-bis, of Code 3, and takes account of population, distances, territory and capacity. As an ordinary criterion, at least one Advanced Workshop is provided for every three or four Circles 1 of Circle 2, unless a different planning is adopted with reasons.

3. The Advanced Workshop may include welding, light forging, machine tools, electronics and printed circuits, large-format 3D printing, laser cutting and the production of complex PPE, within the limits of the applicable STVs and Safety Sheets. The use of the equipment is reserved to persons with the required training and authorisation. Access may be booked by the Circles 1 of Circle 2; the inventory and the management are federated.

4. The Distributed Micro-stations include, by way of example, common tool cabinets, tool libraries, small 3D printers, sewing stations and disassembly and reuse centres. They are registered with the Production Council and operate in accordance with STVs and Safety Sheets appropriate to their activity.

5. The collection of data is permitted exclusively for safety, the material balance, maintenance and the verification of the Basket. The data are aggregated by workshop, by Circle 1 or by Circle 2. Individual behavioural profiling is prohibited. Retention periods and accesses are minimised and defined in the STVs, in conformity with Code 7, Article 2.2.

6. Stocks of materials are sized in relation to the Basket and to the needs of levels 1–3 and may not be built up for speculative purposes. No Circle may be organised in such a way as to depend structurally on a single other Circle for the Material Basket.

Art. 7 — Safety, PPE and training

1. No activity in a workshop may be carried out without a Safety Sheet in force. The Production Council prepares the technical documentation relating to the machine and the procedure. The validation of the general safety aspects falls to the Safety Council in accordance with Code 14. The Health Council gives the mandatory opinion on the health aspects, on PPE and on the consequences for health. The validation of safety does not transfer the technical competences over production.

2. The PPE necessary for workshop activities is a common good managed in the Health Storehouse in accordance with the competent Codes, assigned free of charge, maintained and traced. The unavailability of an item of PPE prescribed as essential entails the suspension of the activity concerned, not a sanction against the person.

3. The minimum training required is as follows:

   a) L1 to access the Base Workshop and the Micro-stations;

   b) L2 to use the advanced machines and to carry out the activities that the STV classifies as specialist;

   c) L3 to design or validate STVs and to run an Advanced Workshop, within the limits of the competences provided for by Codes 2, 5 and 14.

4. Anyone who detects an immediate risk may interrupt the activity and report it to the competent Council, and may not suffer any detriment for an interruption made in good faith.

5. Digital fabricators are fitted with the open-source and inspectable hardware filter provided for by Article 16.1 of the Constitution, which prevents exclusively the fabrication of weapons and their components. It

is prohibited to fabricate such objects or to hold their designs. Tampering with the filter constitutes a Grade 4 offence.

Art. 8 — Register of material Dependencies

1. The Production Council feeds the Dependency Register provided for by Code 15, distinguishing at least:

   a) **material dependency**: dependency on external raw materials necessary for the Basket;

   b) **technological dependency**: dependency on proprietary machines, closed firmware, tools not repairable locally or imported critical components;

   c) **logistical dependency**: unavailability within the Confederation of materials or spare parts that cannot be regenerated locally.

2. Every critical dependency is accompanied by an exit plan with a person responsible, a fixed final deadline, a verifiable objective and continuity measures. Dependencies that affect the Basket or PPE have priority.

3. At least one annual federated audit on material autonomy is carried out by independent L3 members drawn by lot in accordance with Article 36 of Code 4.

Art. 9 — Prohibitions and guarantees

1. It is prohibited to:

   a) derive rent from the essential tools of the Basket or from the use of the Base Workshop;

   b) impose paywalls on the STVs of life-saving objects and on essential PPE;

   c) claim exclusive rights or patents over the essential knowledge necessary for the production of the Basket;

   d) accumulate materials or tools individually beyond the technical limit resolved by the Assembly where this compromises the availability of the Basket or the balance between the workshops;

   e) refuse the repair of a good of the Basket as a sanctioning measure;

   f) transfer control of the Base Workshop or of the Federated Advanced Workshop to parties external to the Confederation.

2. Violations are established and dealt with in accordance with Code 4. Suspension or deactivation measures are proportionate and reasoned and, where technically possible, ensure the continuity of levels 1–3; in any case the continuity of levels 1 and 2 is guaranteed, unless continuation entails a serious and immediate risk.

3. A machine may be switched off completely where this is necessary to prevent further harm. For critical or vital functions, the suspension is accompanied, where technically possible, by a continuity procedure or by a conforming substitute function.

4. Appeal to the Guarantee is permitted within fourteen days in accordance with the procedures of Code 4 for procedural defects, manifest technical unreasonableness, discrimination or disproportion.

Art. 10 — Transitional provisions and coordination

1. This Annex is coordinated with:

   a) Code 1, Articles 18 and 19, for access to and scarcity of the Basket, and Article 32 for the hierarchy of priorities;

   b) Code 2 for STVs and quality validation;

   c) Code 3 for resolutions;

   d) Code 4 for the Guarantee and remedies;

   e) Code 5 for L2 and L3 competences;

   f) Code 6 for PPE and health aspects;

g) Code 7 for the Network and data processing;

h) Code 8 for energy and the microgrid;

i) Codes 12 and 14 for frailties, general safety and Safety Sheets;

j) Code 15 for the Dependency Register.

2. Within six months of entry into force, every Circle 1:

a) inventories existing tools and equipment;

b) activates a minimum Base Workshop, with a provisional STV and a Safety Sheet conforming to the applicable transitional rules;

c) guarantees the minimum Material Basket, including by means of reuse, provided that it is safe and conforming.

3. Within twelve months of entry into force, in every Circle 2 the shared Production Council provided for by Article 12-bis, paragraph 6-bis, of Code 3 activates at least one minimum Advanced Workshop in accordance with Article 6, paragraph 2, a federated inventory and a node of the Network of workshops, in coordination with the transition provided for by Code 8. If Circle 2 is constituted after entry into force, the time limit runs from its constitution. For documented overlaps of the transitional burdens of Codes 8 and 9, the General Assembly of Circle 2 may grant a reasoned extension not exceeding six months, with a binding plan and a guarantee of the continuity of the Basket.

4. Within twenty-four months of entry into force, an exit plan is adopted for the critical material dependencies relating to the Basket and to PPE, with a fixed final deadline for each dependency.

5. The provisional validation of production competences is governed by Code 5, Article 10, paragraph 2. The time limits stated in this Annex are six months for L3 competences and twelve months for L2 competences, unless Code 5, which prevails, provides otherwise.

6. Machines, functions and methods of management that do not conform to Article 9 are brought into conformity or deactivated within ninety days, in accordance with Code 4. Where an immediate deactivation would compromise a vital function of level 1 or 2, the Assembly approves within the same time limit a binding plan for substitution or progressive deactivation, with a fixed deadline and guaranteed continuity. The immediate suspension of an activity that presents a serious and immediate risk remains mandatory.